Kinetic energy

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Calculate the kinetic energy of the plane.

- $E_k = 0.5 \times 9000 \times 100^2$
- $E_k = 45\,000\,000$ J (45 MJ)

2a) Calculate the kinetic energy of each driver.

- Driver 1: $E_k = 0.5 \times 50 \times 40^2 = 40\,000$ J (40 kJ)
- Driver 2: $E_k = 0.5 \times 70 \times 40^2 = 56\,000$ J (56 kJ)

2b) Explain the difference.

- The second driver has more kinetic energy because their mass is larger (E_k is proportional to mass).

3 A crane raises a 200 kg load 30 m in one minute. Calculate the kinetic energy of the load while it is moving.

- $v = d \div t = 30 \div 60 = 0.5$ m/s
- $E_k = 0.5 \times 200 \times 0.5^2 = 25$ J

4 A racing car travels at a constant 50 m/s and has 500 kJ of kinetic energy. Calculate its mass.

- $m = 2E_k \div v^2 = 2 \times 500\,000 \div 50^2$
- $m = 400$ kg

5 A runner crosses the line at 7 m/s with 1862 J of kinetic energy. Calculate the runner's mass.

- $m = 2 \times 1862 \div 7^2 = 76$ kg

6 A remote-controlled car travels at 1.2 m/s with 0.216 J of kinetic energy. Calculate its mass.

- $m = 2 \times 0.216 \div 1.2^2 = 0.30$ kg

7 A 2800 kg lorry has 132 kJ of kinetic energy. Calculate its speed.

- $v^2 = 2E_k \div m = 2 \times 132\,000 \div 2800 = 94.3$
- $v = 9.7$ m/s

8 A 250 kg motorbike travels at 18 m/s and a 950 kg car at 12.5 m/s. Which has more kinetic energy? Show your working.

- Motorbike: $0.5 \times 250 \times 18^2 = 40\,500$ J = 41 kJ
- Car: $0.5 \times 950 \times 12.5^2 = 74\,219$ J = 74 kJ
- The car has more kinetic energy.

9 An 850 kg car slows from 20 m/s to 12 m/s. Calculate the change in its kinetic energy.

- Before: $0.5 \times 850 \times 20^2 = 170\,000$ J
- After: $0.5 \times 850 \times 12^2 = 61\,200$ J
- Change = $170\,000 - 61\,200 = 108\,800$ J = 110 kJ

10a) Calculate the kinetic energy at the lowest point, where the velocity is 15 m/s.

- $E_k = 0.5 \times 2500 \times 15^2 = 281\,250$ J = 280 kJ

10b) Calculate the maximum height the swing reaches, assuming no energy is lost. (Use $E_p = mgh$.)

- $E_p = E_k$, so $mgh = 281\,250$
- $h = 281\,250 \div (2500 \times 9.8)$
- $h = 11$ m

10c) State one cause of energy loss in the swing.

- Air resistance, or friction at the pivot (bar) and ropes.

10d) State how this energy loss could be reduced.

- Streamline the swing to reduce air resistance, or lubricate the pivot to reduce friction.